ORGANIC INTELLIGENCE PROTOCOL

OIP v1.0.40

Consolidated Research-Stage Technical Master Reference and Finalization Record

DOCUMENT EDITION	1.0 | 05 November 2026
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DOCUMENT STATUS	ADMINISTRATIVELY LOCKED RESEARCH SNAPSHOT
PROTOCOL STATUS	NOT APPROVED | NOT SCIENTIFICALLY VALIDATED | NOT FROZEN
PRIMARY EVIDENCE PACKAGE	12 original source files and 119 reconciled artifact-baseline records

Evidence-first. Fail-closed. Explicit about what is known, selected, pending, and not established.

This is a controlled status and technical reference document based on the supplied source pack and the reconciled audit export. It records the state reached as of 05 November 2026. It does not issue a new OIP protocol version or authorize a production, empirical, scientific, or commercial performance claim.

PREPARED FOR TECHNICAL REVIEW, SOURCE RECONCILIATION, AND CONTROLLED FUTURE VALIDATION

Document map

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Status convention

PASS refers only to the defined artifact check or test scope. PENDING means a required decision has not been recorded as approved. NOT ESTABLISHED means the current evidence is insufficient. NOT AUTHORIZED means the current governance record does not permit the activity. OPEN means the evidence or decision remains unresolved.

01. Executive determination

What is complete, what is locked, and what is still not authorized.

RESEARCH SNAPSHOT	LOCKED AS AN ADMINISTRATIVE EVIDENCE BASELINE
PROTOCOL FREEZE	NOT READY / NOT AUTHORIZED
SCIENTIFIC VALIDATION	NOT ESTABLISHED

The current package is sufficiently assembled to preserve and communicate the research-stage record. The source archive is internally reconciled to its inventory, the current artifact snapshot is hash-verifiable, mathematical test outputs are recorded, and the latest proposal registry identifies UC001-UC004 consistently. These are material process achievements.

They do not establish that OIP constructs are validly measured, that model weights are empirically calibrated, that OIP outperforms a comparator, or that the candidate engine is approved for production. The package itself says release is not authorized. This document preserves that boundary.

Decision taken for this document

Lock the present research snapshot, not the protocol. Preserve the current source pack, engine candidate, specification draft, audit reports, proposals, old manifest, new manifest, and this report as a dated evidence set. Do not rewrite historical records to make them appear consistent. Any later correction should be made in a new, versioned package with a change log and new hashes.

Current status in one view

Area	Evidence-bounded status
Source archive	12/12 source files match recorded sizes and SHA-256 values in the supplied inventory.
Reconciled artifact baseline	119 records; 119/119 verified within the new snapshot.
Mathematical test suite	11/11 recorded cases passed; synthetic arithmetic fixtures only.
Source-compliance review	10/10 reviewed checks passed; not a complete semantic re-audit.
Candidate engine regression	44/44 reported pass for engine SHA shown in this record; no scientific validation.
Historical defect closure	9 retest passes reported; 0 fully closed; HK009 remains open.
Owner / contract approvals	Four UC proposals remain pending approval; contract is candidate, not approved.
Measurement / scoring	Not authorized for all 14 measured variables in the authorization audit.

The strongest defensible claim is that a research-stage mathematical and software audit package has been assembled and its current exported artifacts have a reproducible hash baseline. No stronger claim is made.

02. Administrative snapshot lock

A precise distinction between freezing a record and approving a protocol.

Lock identifier: OIP-LOCK-2026-11-05-01
Effective date: 05 November 2026
Scope: Current OIP_MATHEMATICAL_COMPLETION_01 research-stage export, its 12-file original source archive, associated separate Finalization Decision Register notebook, and this report.

What this lock means

- The current evidence package is preserved as a dated baseline.
- No existing source file, audit result, formula record, or historical discrepancy is silently overwritten.
- The reconciled ZIP hash and original-source ZIP hash identify the supplied snapshots.
- Future work starts from a copy or a new version, with a new manifest and change log.

What this lock does not mean

- It is not a normative OIP specification freeze.
- It does not approve confirmatory implementation of Score or UC001-UC004.
- It does not grant measurement or dataset-scoring authorization.
- It does not establish empirical, construct, causal, predictive, or scientific validity.
- It does not authorize production deployment or commercial performance claims.
- It does not establish legal ownership, freedom to operate, or licensing rights.

Administrative lock rationale

This is the correct current decision because the package is stable enough to preserve, while several approval and evidence gates remain open. Continuing to edit the same baseline would weaken the audit trail. Declaring the protocol frozen would overstate the evidence. A dated administrative lock preserves progress without implying scientific approval. This is a documented governance decision, not a storage-level write-protection mechanism. Preserve the exact ZIP unchanged and use its SHA-256 to verify future copies.

Release statement

STATUS: RESEARCH SNAPSHOT LOCKED. PROTOCOL FREEZE: NOT AUTHORIZED.

03. Scope and evidence base

The document is limited to the supplied artifacts and their recorded outputs.

The current review used two uploaded ZIP archives, the separately uploaded Finalization Decision Register notebook, and the structured JSON/TXT/Python artifacts inside the reconciled research export. The original source archive contains six PDFs and six notebooks. The reconciled audit archive contains JSON reports, source-extraction text, specification records, a Python candidate engine, and related runtime artefacts. It contains no .ipynb file.

Source lineage represented in the package

Role	Version / artifact	Evidence boundary
Foundational specification	OIP v1.0.26	Source identity of the supplied foundational PDF is not established by metadata; authoritative artifact still required.
Mathematical provenance	OIP v1.0.27 Addendum	Initial coefficients and operational gaps; explicitly not empirical validation.
Preliminary evaluation	OIP v1.0.28	Historical HR benchmark report; retrospective preliminary evidence only.
Normative methodology	OIP v1.0.30	Construct measurement and dataset qualification protocol; still subject to v1.0.26 source lock.
Dataset evidence boundaries	OIP v1.0.34	Synthesis of four dataset audit records; does not add new empirical validation.
Consolidated reference	OIP v1.0.40	Current consolidated master reference; retains open evidence and claim boundaries.

What the current package does not contain

- The raw empirical datasets required to reproduce historical dataset results.
- A verified authoritative v1.0.26 primary PDF artifact that resolves the source-identity question.
- A complete independent reproduction of the prior empirical or dataset audits.
- Owner-approved UC001-UC004 contract decisions.
- An exact copy of the current Kaggle working notebook inside the Reconciled ZIP.

The separate uploaded oip-finalization-decision-register-01.ipynb has 163 cells and is identified by its own SHA-256 in the package manifest section. It is not treated as proof that the current Kaggle working notebook is archived in the ZIP.

04. Framework architecture

Documented architecture is not equivalent to operationally complete constructs.

```
OIP = HDAI Layer + CGW Layer + Arbitration Layer
```

The source-aligned architecture combines High-Data Analytical Intelligence (HDAI) with Contextual Grounded Wisdom (CGW), governed through an Arbitration Layer. The consolidated reference identifies three key pillars: Grounded Feedback Loop (GFL), Infrastructure Independence (II), and Adaptive Moral Logic (AML).

Pillar	Current description in reviewed package	Operational state
GFL	Grounded Feedback Loop; source description includes real-world outcome correction.	INCOMPLETE
II	Infrastructure Independence, derived as 1 - IDS.	DERIVED FROM IDS; IDS measurement incomplete
AML	Adaptive Moral Logic; source description includes life-preserving prioritization.	INCOMPLETE

Interpretation boundary

The architecture is a documented conceptual framework. Current artifacts do not establish that each pillar can be measured reliably, that the pillars are independent or exhaustive, or that the combined system produces better decisions. Those are research questions for a later measurement and validation phase.

05. Mathematical reference register

Expressions retained from source lineage and the owner decision register. This is a reference record, not an authorization to score data.

ID	Name	Expression	Boundary / scope	Status
F01	Infrastructure Independence	II = 1 - IDS	IDS in [0,1]; IDS=0 gives II=1.	Source-supported expression; IDS measurement incomplete.
F02	Living Intelligence Value	LIV = (RI * SR * AR) / IDS	IDS=0 -> LIV=UNDEFINED. No epsilon, clipping, or denominator substitution.	Inputs and measurement protocols incomplete.
F03	OIP Stability	OIP Stability = GFL * (1 - IDS) * AML	IDS=0 alone does not make this undefined if GFL and AML are defined.	GFL, IDS, AML measurement incomplete.
F04	Neutral Competence Score	Score = 0.6*OE + 0.3*CV + 0.1*VIS	Additive reference form selected; historical alternative kept separate.	Confirmatory implementation pending; weights not empirically calibrated.
F05	Survival Intelligence Score	SIS = 0.5*RC + 0.3*AFL + 0.2*CAS	Distinct from OIP Stability.	Three-term reference form selected; component measurement incomplete.
F06	Stress-Response Intelligence Score	IS = 0.1*SD + 0.9*CW	Situational model; not a universal constant.	Measurement and applicable conditions incomplete.
F07	Historical Score representation	Score = 0.6*Time*Variable*Stability	Preserved as historical notation; not selected for v1.0.40 reference execution.	Earlier primary-source locator unresolved.

Calibration warning

OIP v1.0.27 characterizes Score, SIS, and stress-model weights as initial calibration parameters. They are recorded model choices, not empirically derived or validated constants. Their use for dataset-level scoring remains unauthorized.

06. Formula lineage and unresolved source conflicts

The report preserves historical discrepancies instead of silently reconciling them.

Score expression

The additive expression is recorded in the supplied foundational PDF at pages 1 and 6, in OIP v1.0.30 at page 10, and in OIP v1.0.40 at page 5. OIP v1.0.30 page 9 also reports the historical notation Score = 0.6 * Time * Variable * Stability. The earlier primary-source location for that alternative has not been established.

Current disposition: additive reference form selected; confirmatory implementation remains pending. Historical notation retained in the archive and excluded from current candidate execution.

SIS expression

The three-term reference expression is SIS = 0.5 * RC + 0.3 * AFL + 0.2 * CAS. The source review records an earlier summary omission of a term. The three-term form is selected in the owner decision register; the historical difference is preserved, not rewritten.

Terminology

- RC appears as both Resilience Capacity and Resilience Coefficient in the source lineage.
- AFL appears as Adaptive Feedback Learning and Adaptive Feedback Loop; earlier lineage also uses Adaptive Flexibility.
- v1.0.40 is selected as the canonical terminology reference in the recorded owner decision, while historic variants remain preserved.

Source authority

The supplied foundational PDF has a matching hash against the internal source inventory, but its metadata/title does not establish that it is the authoritative v1.0.26 artifact named by later documents. Hash matching proves that the supplied file is the file listed in the inventory; it does not prove the inventory itself is the authoritative primary-source baseline.

07. Variables and measurement authorization

Documented descriptions, where found, must not be mistaken for operational measurement protocols.

The authorization audit records 14 measured constructs/inputs. For every one, measurement authorization is NOT_GRANTED and dataset-scoring authorization is NOT_GRANTED. The authorization audit explicitly preserves this boundary.

Variable group	Variables	Recorded state
Source-described constructs	GFL, IDS, AML, RI, SR, AR, CAS, SD, CW	Some documentary descriptions located; measurement protocol incomplete or unresolved.
Definitions not located in reviewed v1.0.30 pp. 5-12	OE, CV, VIS, RC, AFL	Limited page-review finding only; not proof that no definition exists anywhere in the source pack.
Terminology requiring reconciliation	RC, AFL	Canonical v1.0.40 labels selected; historical wording preserved.

Source-derived descriptions recorded in reviewed v1.0.30 pages

- GFL: real-world outcome correction; source expressions include a function of real-world outcome error correction rate and feedback frequency multiplied by behavioral update speed.
- IDS: degree to which reasoning requires external systems; source range [0,1].
- AML: life-preserving prioritization; detailed source expression refers to life-preservation weight relative to prestige-optimization weight.
- RI, SR, AR: Resilience Impact, Scalability Range, and Adaptation Rate; measurement protocols unresolved.
- CAS: Contextual Adaptation Speed; operational protocol unresolved.
- SD/CW: abbreviations used in the situational stress-response formula; measurement protocol unresolved.

No proxy variables, imputation rules, or unsupported measurement conversions have been introduced to complete these gaps.

08. Recorded owner decision register

The supplied register contains 12 recorded choices. Its overall status is pending verification; it is not the same as the 12 unresolved issue register.

ID	Topic	Recorded decision	Limitation / status
OD001	Score	Select additive reference formula.	Reference form selected; not empirically calibrated or validated.
OD002	Historical Score	Do not implement historical alternative notation.	Archival discrepancy retained; original locator unresolved.
OD003	SIS	Use three-term reference formula.	Component measurement needs separate authorization.
OD004	II	Retain II = 1 - IDS.	IDS must be within [0,1]; measurement incomplete.
OD005	OIP Stability	Retain GFL*(1-IDS)*AML.	Construct measurement not authorized.
OD006	LIV	Retain (RI*SR*AR)/IDS and IDS=0 boundary.	LIV undefined at IDS=0; no epsilon or substitution.
OD007	IS	Keep IS = 0.1*SD + 0.9*CW as situational.	Not a universal constant.
OD008	Terminology	Use v1.0.40 terms for RC, AFL, CAS.	Historical variants preserved.
OD009	Missingness	Never silently convert missing to zero.	Mandatory fail-closed rule.
OD010	Historical datasets	Keep unresolved dataset audits outside scoring authorization.	Historical evidence only.
OD011	Measurement	Require evidence-based authorization before scoring.	No invented proxy or unsupported transform.
OD012	Validation	Do not claim scientific/empirical validation without independent evidence.	Validation not established.

The register is marked OWNER_DECISIONS_RECORDED_PENDING_VERIFICATION. This report records the entries as they appear in the supplied file. It does not alter their source status or claim that all owner decisions have received independent verification.

09. Mathematical and source-compliance test evidence

PASS applies only to the cases and checks recorded in the submitted artifacts.

Test set	Recorded outcome	Meaning
Mathematical cases	11/11 PASS	Deterministic arithmetic and fail-closed cases; synthetic fixtures, not empirical evidence.
Source-compliance review	10/10 PASS	Ten specified consistency and claim-boundary checks; not a complete semantic source re-audit.
Runtime regression	44/44 PASS	Candidate engine regression report; engine hash identified; owner approval remains pending.

Key mathematical boundary checks

- At IDS=0: LIV is UNDEFINED; II=1; OIP Stability can remain defined if GFL and AML are defined.
- Historical multiplicative Score notation remains excluded from execution.
- Missing IDS is not coerced to zero; missing input is blocked.
- IDS fixtures are constrained to [0,1].
- SIS and OIP Stability remain distinct expressions.

What these tests do not establish

They do not establish construct validity, measurement reliability, empirical calibration, out-of-sample performance, causal effects, predictive validity, generalizability, safety, or commercial benefit. The mathematical test report itself limits its conclusion to the deterministic cases executed. The compliance review states that it is not a complete semantic re-audit of all source PDFs.

10. Candidate engine and implementation contract

The candidate engine is identifiable and regression-tested in the saved record, but it is not an approved release.

Engine file: engine/oip_candidate_engine.py
SHA-256: `6ad869e03d27bb407c2ebe9502996a8d71f3074445e2272ac9715ff296413610`
Release status: CANDIDATE_NOT_APPROVED

The final audit manifest records the engine hash and reports 44 of 44 runtime regression tests as passed. The regression report itself records the same engine hash and lists cases covering numeric input types, IDS boundaries, formula arithmetic, overflow handling, pipeline input validation, output contract, and hashing repeatability.

UC001-UC004 proposal registry

ID	Proposal	Identity	Decision status	Owner approval	Current SHA-256
UC001	oip_uc001_status_semantics_proposal.json	IDENTITY VALID	PROPOSED_NOT_APPROVED	PENDING	dfa0b45cd4897c152fb3f7532b7dc01410d42b751367b1e5f5cc16ac1d1ffd96
UC002	oip_uc002_numeric_type_policy_proposal.json	IDENTITY VALID	PROPOSED_NOT_APPROVED	PENDING	38398c7d35d3c3090e29258f5e3e4f2f81a998271f1ba39711b503e124fecd8d
UC003	oip_uc003_output_schema_proposal.json	IDENTITY VALID	PROPOSED_NOT_APPROVED	PENDING	a3509ef19a78b54b5baa142878c629dcacd5a9239456bb57c14c8c2b03820a82
UC004	oip_uc004_canonical_hashing_proposal.json	IDENTITY VALID	PROPOSED_NOT_APPROVED	PENDING	58da4cd96ef80cb9f2b64f360f07fbbf93ba4838cf6f5969401577e32aa1cb14

The later canonical registry uses decision_id and reports all four identities valid, with all owner approvals pending. Cell 139 contains an earlier exact-proposal scan based on a different field and reported no exact records; later reconciliation corrected the identity field and produced the canonical registry. The later registry and Cell 148 consistency report are the controlling evidence for proposal identity, not the earlier diagnostic scan.

The deterministic engine contract remains CANDIDATE_NOT_APPROVED. UC001-UC004 must receive explicit owner decisions before a software release freeze.

11. Historical software defect register

Retest evidence is not the same as formal defect closure.

Measure	Recorded result
Historical defect count	10 (HK001-HK010)
Retest-pass reports	9
Owner decision pending	HK009: status contract mismatch
Fully closed count	0
Scientific validation	NOT_ESTABLISHED
Freeze authorization	False

Disposition summary

- HK001: Score fixture defect - retest pass reported.
- HK002: Non-dictionary pipeline input - retested in Cell 136.
- HK003: Unknown pipeline keys - retested in Cell 136.
- HK004-HK008: overflow, fixture, numeric robustness, and weight validation - retest pass reported.
- HK009: status contract mismatch - observed behavior; owner decision remains pending.
- HK010: validation precedence - retest pass reported.

The closure register explicitly marks every defect fully_resolved: false and gives fully_closed_count: 0. Therefore this report does not mark any of HK001-HK010 closed. It records reported retest outcomes and preserves the open authorization/closure boundary.

12. Historical empirical and dataset evidence boundaries

Prior source reports are recorded as historical claims, not revalidated results from this export review.

OIP v1.0.28 preliminary HR evaluation

The supplied one-page v1.0.28 report states an open-source HR benchmark cohort of N=54,808 and reports a promotion-yield comparison of 12.66% for a baseline versus 20.62% for the OIP v1.0.28 vector, an absolute difference of 7.96 percentage points, with a reported 95% confidence interval of [6.57%, 9.34%] and p=3.59 x 10^-29. The report labels the result retrospective preliminary empirical evidence and does not establish causal efficacy or universal applicability. The raw dataset and a complete independent reproduction are not present in the current source export; this report does not endorse the figures as independently validated.

OIP v1.0.34 four-dataset synthesis

OIP v1.0.34 describes a synthesis of four dataset audit records: BEMP, Malawi IHPS, Uganda UNPS 2019/20, and Ethiopia ESS4. It states that downstream conditions for authorized construct operationalization and empirical execution were not established in the available records, and that the response was to withhold authorization rather than use unsupported proxies, silent imputation, synthetic outcomes, or post-hoc scoring.

Waymo and other v1.0.40 open evidence

The current master reference retains an open requirement to obtain the complete Waymo primary audit package and reproduce the reported counts. It also retains separate open evidence items for BEMP, Malawi, Tanzania NPS, Nigeria GHS, KHDS, and OIP v0.2.0 test/repository history. No new result is inferred from the presence of these open-item records.

No dataset scoring is authorized by the current package. Historical audit outputs must not be presented as current OIP validation.

13. Twelve mathematical and operational open issues

These are separate from the twelve recorded owner choices in Section 08.

Issue	Open item	Current status	Required next action
ISSUE_001	Historical multiplicative Score primary-source locator	UNRESOLVED	Identify and verify the original source and its version history, or retain the locator as unresolved.
ISSUE_002	Confirmatory authorization of additive Score	PENDING CONFIRMATORY LOCK	Explicitly approve or reject confirmatory implementation after resolving the historical formula conflict.
ISSUE_003	GFL operational definition and measurement protocol	NOT ESTABLISHED	Define observable inputs, measurement procedure, missingness rules and validation requirements.
ISSUE_004	IDS operational measurement protocol	NOT ESTABLISHED	Specify measurement inputs, calculation rules, boundary handling and validation procedure.
ISSUE_005	AML operational definition and measurement protocol	NOT ESTABLISHED	Define observable components, measurement rules and validation requirements.
ISSUE_006	Operational definitions of Score and SIS components	NOT ESTABLISHED	Specify each component's construct definition, observable indicators, measurement procedure and missingness handling.
ISSUE_007	Operational definitions of LIV components	NOT ESTABLISHED	Define RI, SR and AR independently, including their measurement procedures and evidence requirements.
ISSUE_008	RC terminology conflict	UNRESOLVED	Select and document the canonical term, or explicitly distinguish separate constructs if intended.
ISSUE_009	AFL terminology conflict	UNRESOLVED	Select and document the canonical term, or explicitly distinguish separate constructs if intended.
ISSUE_010	Missingness and boundary rules	PARTIALLY SPECIFIED	Specify component-level missingness, invalid-input, non-finite and boundary handling without converting missing values to zero.
ISSUE_011	Previous software audit register hash discrepancy	PRESERVED UNRESOLVED INTEGRITY DISCREPANCY	Investigate the discrepancy using the original artifacts and reproducible hash procedure. Do not infer a root cause.
ISSUE_012	Empirical and scientific validation	NOT ESTABLISHED	Define a pre-registered validation protocol, source data requirements, independent review and acceptance criteria.

The register labels itself DRAFT_PENDING_OWNER_DECISIONS, with 12 issues and 12 owner decisions required. The current report preserves each status exactly as recorded. Some items can be retained as explicitly unresolved rather than “solved” by inference; measurement protocols and confirmatory validation require additional evidence and review.

14. Source conflicts and v1.0.40 master open evidence

The source review tracks 7 source conflicts/gaps and 8 open master evidence items (15 in total).

Source conflicts and gaps - SRC001 to SRC007

ID	Issue	Status	Required action
SRC001	Authoritative v1.0.26 foundational source identity	UNRESOLVED	Obtain and verify the authoritative v1.0.26 primary artifact.
SRC002	Historical Score notation discrepancy	REFERENCE FORM SELECTED ARCHIVAL DISCREPANCY RETAINED	Retain the selected reference form and preserve the unresolved historical locator.
SRC003	Historical SIS summary omission	REFERENCE FORM SELECTED ARCHIVAL DISCREPANCY RETAINED	Preserve the historical difference; do not silently rewrite earlier records.
SRC004	AFL terminology across versions	CANONICAL TERM SELECTED HISTORICAL VARIANT PRESERVED	Preserve version-specific wording and avoid assuming semantic equivalence without evidence.
SRC005	Construct operational measurement protocols	MEASUREMENT NOT AUTHORIZED	Obtain source-backed operational protocols and complete construct-specific evidence review.
SRC006	Definitions not located in reviewed v1.0.30 pages	SOURCE REVIEW INCOMPLETE	Review remaining authoritative source pages and verify exact definitions.
SRC007	v1.0.30 source lock against authoritative v1.0.26	PENDING AUTHORITATIVE SOURCE LOCK	Verify the authoritative v1.0.26 artifact and reconcile the source lock.

Master open evidence - MST001 to MST008

ID	Open evidence item	Status	Source reference
MST001	Reconstruct exact BEMP notebook runtime outcomes	OPEN	v1.0.40 page 10
MST002	Reconcile Malawi file inventory counts and their meanings	OPEN	v1.0.40 page 10
MST003	Verify Tanzania NPS Wave 4 canonical manifest hash	OPEN	v1.0.40 page 10
MST004	Verify Nigeria GHS Wave 5 manifest and Version 01 versus Version 2 identity conflict	OPEN	v1.0.40 page 11
MST005	Re-establish KHDS structural-pair count from source files	OPEN	v1.0.40 page 11
MST006	Obtain complete Waymo primary audit package and reproduce reported counts	OPEN	v1.0.40 page 11
MST007	Reconcile OIP v0.2.0 test scopes and execution records	OPEN	v1.0.40 page 11
MST008	Complete OIP v0.2.0 repository and archive licence review	OPEN	v1.0.40 page 11

The source/open-evidence register explicitly states that unresolved items must not be inferred closed, measurement authorization is not granted, dataset-scoring authorization is not granted, and scientific validation is not established.

15. Recommended validation pathway

The next phase should resolve evidence gaps before measuring performance.

Stage	Required work
Stage A - source authority	Acquire and verify the authoritative v1.0.26 primary artifact. Reconcile source identities, source versions, formula lineage and the normative authority chain. Preserve any unresolved historical locator rather than guessing.
Stage B - specification completion	Complete source-derived definitions and explicit measurement protocols for each construct. Specify observable inputs, units, normalization, validity ranges, missingness, non-finite values, boundary behavior, and decision authority.
Stage C - owner decisions	Review OD001-OD012 and ISSUE_001-ISSUE_012 separately. Resolve or defer each item with rationale, evidence reference, approver, date, and effect on implementation. Approve UC001-UC004 only after reading their full proposals.
Stage D - independent mathematical and software review	Provide the same frozen source pack, specification, engine source, and tests to at least two independent reviewers. Require reproducible observations and a written disagreement log. Re-run tests after any approved change.
Stage E - empirical pilot pre-registration	Before seeing pilot outcomes, freeze dataset provenance, cohort, comparators, outcomes, time split, metrics, missing-data treatment, analysis code, exclusion criteria, and stopping rules.
Stage F - independent reproduction and release gate	Reproduce results from raw sources in a clean environment. Report failures and uncertainty. Release only the narrow claim supported by the evidence; keep scientific, production, and commercial approvals distinct.

Minimum acceptance criteria for a later freeze

- All formulas trace to a verified, authoritative source or are explicitly labelled as owner-selected design choices.
- All operationalized variables have approved measurement protocols and missingness/boundary rules.
- Every test is linked to a requirement and the exact source/code versions.
- All open issues are resolved or formally deferred, with no silent closure.
- Independent reviewers can reproduce key calculations and audit outputs.
- All release claims stay within the evidence actually established.

16. Commercial evaluation and licensing boundary

A company-facing dossier can be prepared, but verified savings and ROI cannot be claimed yet.

OIP may be presented to a prospective organization as a research-stage decision-assessment framework under evaluation. It should not be marketed as proven to reduce costs, improve hiring outcomes, improve safety, increase profit, or outperform existing systems. Those are hypotheses requiring a controlled pilot.

Pilot design that could support a commercial case

Element	Minimum specification
Use case	One bounded decision workflow with a documented current process and an accountable business owner.
Baseline	Measure existing time, review effort, error/rework rates, escalation frequency, and cost per completed case before OIP exposure.
Comparator	Pre-specify the incumbent process or accepted baseline. Use a time split, matched cases, or a controlled comparison appropriate to the workflow.
Measurement	Only use OIP constructs after their measurement protocols and authorization are approved. Do not substitute convenience proxies.
Controls	Human oversight, version/hash logging, missing-data blocking, documented overrides, and stop conditions.
Primary outcomes	Pre-register outcome quality, review time, error/rework, completeness, and operational cost metrics.
Analysis	Report effect sizes, uncertainty intervals, missingness, exceptions, and negative findings; independent reviewer reproduces calculations.

Transparent business case formulas

```
Verified gross savings = baseline process cost - OIP-assisted process cost
```

```
Net benefit = verified gross savings - total OIP implementation and licensing cost
```

```
ROI (%) = (net benefit / total OIP cost) * 100
```

No numeric estimate is inserted here because no prospective pilot baseline, implementation cost, license fee, or measured OIP-assisted outcome has been supplied. A commercial licensing term sheet should remain conditional and subject to legal review, IP provenance review, data-rights review, third-party software licence review, and a defined pilot scope.

17. Release gates and change control

Separate the evidence, specification, software, scientific, and commercial gates.

Gate	Current status	Reason / boundary
Artifact integrity	PASS for supplied current snapshot	Does not establish semantic correctness.
Source authority	OPEN	Authoritative v1.0.26 source identity/source lock unresolved.
Mathematical specification	DRAFT / PENDING REVIEW	Reference formulas captured; operational protocol and confirmatory Score lock remain open.
Implementation contract	CANDIDATE_NOT_APPROVED	UC001-UC004 owner approvals pending.
Historical defect closure	PARTIAL	9 retest pass reports; zero fully closed; HK009 remains open.
Measurement authorization	NOT GRANTED	All 14 variables are not authorized for measurement/dataset scoring.
Scientific validation	NOT ESTABLISHED	Independent empirical/construct validation not available.
Production release	NOT AUTHORIZED	No production-readiness claim.
Commercial readiness	NOT ASSESSED	No pilot ROI or rights clearance established.

Change-control rule

Any later change must be issued as a new dated version. Record: change ID; affected artifact paths; old and new SHA-256; reason; evidence; approver; tests added or rerun; impact on open issues; and release decision. Do not replace the locked 05 November 2026 baseline. A later baseline may supersede it for operational use, but the earlier archive remains immutable.

Recommended next controlled work package

OIP_FINALIZATION_PHASE_02 - source authority verification, owner issue disposition, measurement protocol completion, UC001-UC004 decisions, and independent review preparation. Keep this distinct from the completed audit-package assembly phase.

18. Package provenance and integrity record

Hashes identify the exact supplied files reviewed for this document.

Artifact	SHA-256
Reconciled full audit ZIP	fd681e57057cf42906d6744e4a7e74ff 3c29be215c339a41a3f3f43c1daa5552
Original source ZIP	412287c716c61cb79c505b536c22df73 2e88663c3f0088f353ed31eb4d33c2cf
Current artifact_hash_manifest.json	020ac8a48dbd9afa907689e88bd3c45b 4b1ac924273b24598c0f784f8ee4cbb9
Preserved prior artifact manifest	cfe64702c3279953d1db76e9a13f57a2 3058ec90acfa41dd20eeaff6c9c9a2e5
OIP_FINAL_PACKAGE.json	29a7f118549dfd3ea856b13519e69b2e 1a6698ae78d6385dbcd39a0d2184433a
FULL_AUDIT_EXPORT_MANIFEST.json	b08b7dbae46d821d3ba24757ab36c05a a013f7311dcd889a3c4ad6b4d845a2f7
OIP_FINAL_AUDIT_COMPLETION_MANIFEST.json	87f9c0e8f7be0d55326d4c5f741a5a68 33aff8b3f55d693c2335dec70947b7c8
Candidate engine source	6ad869e03d27bb407c2ebe9502996a8d 71f3074445e2272ac9715ff296413610
Separate Finalization Decision Register notebook	0ce62fb94b1cc25a006eec69eeae1e37 73ea32573a9761e343f887a30510ae32

Snapshot counts and manifest relationships

Record	Value
Previous artifact manifest records	118
Current artifact manifest records	119
Changed records on common paths	26
Added path(s)	artifact_hash_manifest_PRE_RECONCILIATION.json
Removed paths	None
Full export manifest payload file count	120
ZIP entries including directory entry	121
Current baseline verification	119/119 PASS in Cell 152 output; ZIP integrity PASS

Important embedded metadata inconsistency

The file OIP_MATHEMATICAL_COMPLETION_01_FINAL_PACKAGE.json still points to the prior artifact manifest SHA-256 cfe64702c3279953d1db76e9a13f57a23058ec90acfa41dd20eeaff6c9c9a2e5 and record count 118. The current artifact_hash_manifest.json is SHA-256 020ac8a48dbd9afa907689e88bd3c45b4b1ac924273b24598c0f784f8ee4cbb9 with 119 records. The old manifest is preserved. The pointer in the package JSON has not been refreshed, so the export is hash-verifiable but this internal pointer is stale. This is a packaging metadata issue, not evidence that the current 119-file verification failed.

19. Original source inventory

The source archive contains six PDFs and six notebooks. Each file was independently compared with the recorded inventory by size and SHA-256.

Source file	Bytes	SHA-256	Check
OIP_v1_0_27_Addendum.pdf	95,240	fa58b441d9ac15b03781ac6373b23084 c404946d6aebc161f07bdbcb9f93cebd	MATCH
OIP_v1_0_28_Empirical_Report_pdf.pdf	194,354	a9d0d1bb80b90843499641746c4a1e3c 763d35bd3b6f707d60c116d82b1361d3	MATCH
OIP_v1_0_30_Normative_Protocol.pdf	278,304	3565c37c6915572c607ea73b05b80261 d0c34bcb515b5eb2645fef3dbc4a1309	MATCH
OIP_v1_0_34_Four_Dataset_Evidence_Boundary_Report-2.pdf	254,425	cf27d24beafbdabd144180510a3e3df9 ec6552871f4a75184cd1f4ef5d08eb87	MATCH
ORGANIC INTELLIGENCE PROTOCOL (OIP)- A RESILIENCE-WEIGHTED FRAMEWORK FOR BEYOND-INFRASTRUCTURE INTELLIGENCE (2) (1).pdf	255,379	3b82b84e12e9a2eab9e9014f2dd4bcab 5e8880c2215f81d290a06055832a3195	MATCH
Sudharsan_Das_OIP_v1.0.40_Master.pdf	350,039	eceaf45dfdc5e03eff4006b68634cb31 a909795fba4e772b08d9ea0765639c61	MATCH
oip-calculator-kernel-v1-0-0.ipynb	797,026	978a8ea968c7f5400ca7620d47f34baa 894340b6f75022ed9ad73f076dd70879	MATCH
oip-final-mathematical-specification-completion-01.ipynb	202,591	42ad65aafa55c585444da6e8e17d968f 2ea8f42b1b9a616450a650187ef47551	MATCH
oip-master-rebuild-01 (1).ipynb	694,083	248aa90b4c1ff6eaf7f6799d7c7882b6 f0ac3d51442d253dece23c94c2103a9d	MATCH
oip-master-reconciliation-rebuild-01.ipynb	3,479,007	571c7f95451305a7dcf78e367f43b0a5 0704b1102bfa31b05ca8e498fe240262	MATCH
oip-mathematical-specification-and-testing-v1-0-0.ipynb	1,286,806	da3b3fd2aff0602aa5d4ba3c51f48285 8ebf98440a0e04e967b0368416aae574	MATCH
rebuild-02.ipynb	42,577	2ffb54bfb8485aaa7ab0a8e85b4dd566 5a3f095bf0814c00e4cda375f2cdb562	MATCH

SOURCE INVENTORY	12/12 MATCH

The source hashes match the inventory supplied within the audit package. This verifies consistency between the current original-source ZIP and that recorded inventory. It does not establish that the source inventory was anchored to an independent trusted external baseline, nor does it resolve the authoritative identity of the foundational v1.0.26 PDF.

20. Final determination and next instruction

The current package is ready to preserve and communicate with explicit research-stage limitations.

LOCKED: THE 05 NOVEMBER 2026 RESEARCH-STAGE SNAPSHOT.

NOT LOCKED: THE NORMATIVE OIP PROTOCOL, THE SCIENTIFIC CLAIMS, THE MEASUREMENT SYSTEM, OR THE PRODUCTION ENGINE.

What is complete enough to state now

- The original source archive contains 12 inventoried files and all 12 match the supplied inventory hashes and sizes.
- The current audit export has a reproducible SHA-256 baseline with 119/119 artifact checks reported as passing.
- The mathematical and source-compliance test artifacts record 11/11 and 10/10 passes for their bounded test scopes.
- The candidate engine report records 44/44 runtime regression tests passed for the identified candidate engine hash.
- Four software contract proposal identities are registered consistently in the later canonical registry, while owner approval remains pending.
- All major gaps and limitations remain visible in issue, source-evidence, and defect registers.

What must happen later

- Obtain and verify the authoritative v1.0.26 source artifact; review remaining primary-source pages.
- Resolve or formally defer the 12 mathematical/operational issues and 15 source/master evidence items.
- Obtain explicit approval for the four software contract proposals and close the status-contract decision HK009.
- Complete authorized measurement protocols before any construct scoring or empirical evaluation.
- Pre-register a pilot and arrange independent reproduction before making performance, safety, savings, or ROI claims.
- Upload and preserve the exact current Kaggle working notebook separately; the reconciled ZIP itself contains no .ipynb file.

This report is the final status record for the work evidenced in the supplied package as of 05 November 2026. It is not a declaration that all OIP research is complete. It is designed to let a reviewer distinguish completed audit work from selected design decisions, pending approvals, and work that cannot yet be done because the required source data or measurement evidence is absent.

END OF DOCUMENT